

Information Analysis and Visualisation

Laboratory Sessions 6

Statistical Distributions

This laboratory session is focused on Statistical Distributions, more specifically Uniform Distributions and Normal Distributions. We will be using the Python libraries NumPy and pandas, which provide the data structures and the functionality needed for solving statistical problems. We will be using the Matplotlib library to visualise the datasets.

Task 1

Initialise a one-dimensional NumPy array representing a uniform distribution containing 50 values from the interval [0 .. 100].

Task 2

Find the minimum value, the maximum values and the range of the dataset.

Task 3

Initialise a one-dimensional array representing a normal distribution of 1000 data points with mean value 17 and standard deviation 0.2.

Task 4

Find the minimum and the maximum values of the dataset and calculate the range.

Task 5

Visualise the dataset by using a histogram with 10 bins. Visualise the probability density function.

Probability density function f :

$$f = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(X-\mu)^2}{2\sigma^2}}$$

Where X represents the data points, μ is the mean value and σ is the standard deviation